Name_____ Date_____ Period_____

LESSON 9-4 · HOMEWORK PACKET · SHEET 1 OF 2

Substitute and Solve

Maryland Standard 6.EE.C.9 · Learning target – I can substitute a value into an equation to find the matching output.

REMEMBER

- To find the output, **substitute** the input for x.
- Then use **order of operations**: multiply first, then add or subtract.

WORKED EXAMPLE

Use $y = 6x + 4$ to find y when $x = 5$.

Step 1 Substitute: $y = 6 \times 5 + 4$

Step 2 Multiply, then add: $30 + 4 = 34$

Answer **$y = 34$, so (5, 34)**

PRACTICE

show your work · label units

1. Substitute to find y.

a. $y = 4x + 7, x = 3$ _____

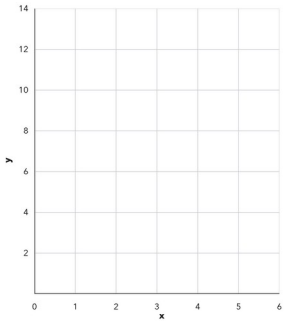
b. $y = 9x, x = 8$ _____

c. $y = 3x - 2, x = 10$ _____

d. $y = x + 15, x = 27$ _____

2. Complete the table for $y = 2x + 3$. Then plot the points and connect them.

x	0	1	2	4	5
y					



3. **Water Tank.** A tank has 20 gallons. It fills at 15 gallons each minute, so $g = 15m + 20$.

a. Gallons after 6 minutes: _____ gal

b. Gallons after 10 minutes: _____ gal

4. **Bike Rental.** A bike rental costs $c = 7h + 5$ dollars for h hours. Find the cost of renting for 3 hours. Then find the cost for 8 hours.

Costs_____

5. **On the Line?** Substitute x to decide whether each point is on the graph of $y = 5x + 2$.

a. (4, 22): $5 \times 4 + 2 =$ ____ On the line? ☐ Yes ☐ No

b. (6, 30): $5 \times 6 + 2 =$ ____ On the line? ☐ Yes ☐ No

6. **Find the Error.** For $y = 4x + 3$ with $x = 5$, Dev added first: $4 + 3 = 7$, and $7 \times 5 = 35$. Explain his mistake and find y.

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Pick a rule like $y = 2x + 1$. Ask someone at home for an input and find the output together.

Name_____ Date_____ Period_____

LESSON 9-4 · HOMEWORK PACKET · SHEET 2 OF 2

Working Backward to a Goal

Maryland Standard 6.EE.C.9 · Learning target – I can work backward from a goal to find the input an equation needs.

REMEMBER

- Know the goal (y)? Substitute it, then solve for x.
- Undo in **reverse order**: subtract the fee first, then divide by the rate.
- In a counting problem, a decimal answer may need to be **rounded up**.

WORKED EXAMPLE

A rental costs $y = 4x + 10$ for x hours. How many hours cost \$46?

Step 1 Subtract 10: $46 - 10 = 36$

Step 2 Divide by 4: $36 \div 4 = 9$

Answer **9 hours**

PRACTICE

show your work · label units

1. Solve for x.

a. $7x = 63$ _____

b. $8x = 104$ _____

c. $3x + 5 = 32$ _____

d. $6x + 12 = 60$ _____

2. **Trip Fund.** A class has \$45 and earns \$15 a week: $y = 15x + 45$. How many weeks until it has \$210?

Answer_____ weeks

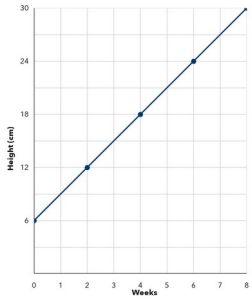
3. **Gift Fund.** A club has \$30 in its gift fund. Members add \$12 each week, so $y = 12x + 30$. How many weeks until the fund has \$150?

Answer_____ weeks

4. **Plant Growth.** A plant is 6 cm tall and grows 3 cm a week: $h = 3w + 6$.

a. Use the graph. When is it 18 cm? _____

b. Use the equation. When is it 33 cm? _____



5. **Round Up.** A box holds 12 cookies. Ana needs 150 cookies, so she says 12.5 boxes. How many boxes should she buy? Explain.

_____

_____

TEST PRACTICE

choose the best answer

6. A rental costs $y = 8x + 12$ dollars for x hours. Max has \$60. How many hours can he rent?

A 5 hours **B** 6 hours **C** 7.5 hours **D** 9 hours

7. Mia solved $5x + 20 = 85$ by dividing 85 by 5 first. What is the correct value of x?

A $x = 17$ **B** $x = 21$ **C** $x = 13$ **D** $x = 65$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___